

Representative Independent Analytical Project: Healthcare Billing & Operational Data Analysis

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Role: Healthcare Operations & Data Analytics **Toolstack:** Python, SQLite (SQL), Pandas, OpenPyXL (Excel Data Modeling) **Document Purpose:** Demonstration of SQL relational database querying, Common Table Expressions (CTEs), dynamic data discretization, and cross-tabulation analysis applied to healthcare billing and reimbursement datasets.

```
import pandas as pd
import sqlite3
import numpy as np

# Load the uploaded file from Colab storage
df_billing = pd.read_csv("healthcare_dataset.csv")

# Clean column names for SQL compatibility
df_billing.columns = df_billing.columns.str.strip().str.replace(' ', '_').str.lower()
# Format dates and billing amounts
df_billing['date_of_admission'] = pd.to_datetime(df_billing['date_of_admission'])
df_billing['discharge_date'] = pd.to_datetime(df_billing['discharge_date'])
df_billing['billing_amount'] = pd.to_numeric(df_billing['billing_amount'], errors='coerce').round(2)

# Calculate length of stay (days)
df_billing['length_of_stay'] = (df_billing['discharge_date'] - df_billing['date_of_admission']).dt.days

# Load into SQLite database in Colab
conn = sqlite3.connect(':memory:')
df_billing.to_sql('billing_claims', conn, index=False, if_exists='replace')

print("Project Author: Christine Nabisswa")
print(f"Successfully loaded {len(df_billing):,} billing claims into SQL database for analysis.")
df_billing[['name', 'medical_condition', 'insurance_provider', 'billing_amount', 'admission_type', 'length_of_stay']].head(5)
```

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Successfully loaded 55,500 billing claims into SQL database for analysis.

	name	medical_condition	insurance_provider	billing_amount	admission_type	length_of_stay	
0	Bobby JacksOn	Cancer	Blue Cross	18,856.28	Urgent	2	
1	LesLie TErRy	Obesity	Medicare	33,643.33	Emergency	6	
2	DaNnY sMitH	Obesity	Aetna	27,955.10	Emergency	15	
3	andrEw waTtS	Diabetes	Medicare	37,909.78	Elective	30	
4	adrIENNE bEll	Cancer	Aetna	14,238.32	Urgent	20	

```
# SQL Query: Revenue share, average stays, and payer ranking
pd.options.display.float_format = '{:,.2f}'.format
```

```
sql_query = """
WITH Payer_Summary AS (
    SELECT
        insurance_provider,
        COUNT(*) AS total_claims,
        ROUND(SUM(billing_amount), 2) AS total_billed,
        ROUND(AVG(billing_amount), 2) AS avg_claim_amount,
        ROUND(AVG(length_of_stay), 1) AS avg_days_stay
    FROM billing_claims
    GROUP BY insurance_provider
)
SELECT
    insurance_provider,
    total_claims,
    total_billed,
    avg_claim_amount,
    avg_days_stay,
    RANK() OVER (ORDER BY total_billed DESC) AS revenue_rank
FROM Payer_Summary
ORDER BY revenue_rank;
"""
```

```
df_payer_summary = pd.read_sql_query(sql_query, conn)
print("--- SQL OUTPUT: Payer Billing Performance ---")
print(df_payer_summary.to_string(index=False))
```

--- SQL OUTPUT: Payer Billing Performance ---

insurance_provider	total_claims	total_billed	avg_claim_amount	avg_days_stay	revenue_rank
Cigna	11249	287,139,345.16	25,525.77	15.50	1
Medicare	11154	285,720,757.70	25,615.99	15.60	2
Blue Cross	11059	283,254,294.21	25,613.01	15.50	3
UnitedHealthcare	11125	282,454,542.59	25,389.17	15.50	4
Aetna	10913	278,863,102.29	25,553.29	15.40	5

```
import warnings
```

```
warnings.filterwarnings('ignore')
```

```
# 1. Excel Dynamic Binning
```

```
bins = [0, 10000, 25000, 40000, np.inf]
```

```
labels = ['Low ($0-$10k)', 'Moderate ($10k-$25k)', 'High ($25k-$40k)', 'Extreme (> $40k)']
```

```
df_billing['billing_tier'] = pd.cut(df_billing['billing_amount'], bins=bins, labels=labels)
```

```
# 2. Pivot Table Aggregation
```

```
excel_pivot = df_billing.pivot_table(
```

```
    index='medical_condition',
```

```
    columns='billing_tier',
```

```
    values='billing_amount',
```

```
    aggfunc=['count', 'mean'],
```

```
    fill_value=0,
```

```
    observed=False
```

```
)
```

```
# 3. Clean Currency Formatting & Render Clean Table Output
```

```
excel_pivot = excel_pivot.round(2)
```

```
print("--- EXCEL MODEL OUTPUT: Condition vs. Billing Tier Summary ---")
```

```
print(excel_pivot.to_string())
```

--- EXCEL MODEL OUTPUT: Condition vs. Billing Tier Summary ---

billing_tier	count				mean		
	Low (\$0-\$10k)	Moderate (\$10k-\$25k)	High (\$25k-\$40k)	Extreme (> \$40k)	Low (\$0-\$10k)	Moderate (\$10k-\$25k)	High
medical_condition							
Arthritis	1745	2807	2843	1902	5,539.97		17,511.95
Asthma	1647	2800	2787	1933	5,456.13		17,471.14
Cancer	1757	2851	2750	1850	5,530.00		17,574.35
Diabetes	1674	2837	2851	1922	5,606.35		17,554.58
Hypertension	1697	2849	2728	1950	5,491.51		17,500.56
Obesity	1624	2754	2961	1873	5,512.82		17,607.39

